

Student ID (NOT your name):

Final Examination: QUESTION BOOKLET

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- Do *NOT* open this question booklet until you are told to do so.
- Write your Student ID number (**NOT** your name) at the top of this page.
- Write your solutions in this booklet.
- No electronic devices are allowed during the exam.
- Be neat! If we can't read it, we can't grade it.
- You can treat any results from lecture or homework as "known," and use them in your work without rederiving them, but do make clear what result you're using. You do not need to explicitly check regularity conditions for the theorems from class that required them.
- For a multi-part problem, you may treat the results of previous parts as given (if you don't prove the result for part (a), you can still use it to solve part (b)).
- I have starred some parts which I believe are the most difficult, and which I expect most students won't necessarily be able to solve in the time allotted. They are generally not worth more points than the less difficult parts, so don't waste too much time on them until you're happy with your answers to the latter.
- Be careful to justify your reasoning and answers. We are primarily interested in your understanding of concepts, so show us what you know.

Good luck!

1. The absent minded philosopher (20 points, 5 points / part).

Your friend in the Philosophy department pays tens of thousands of dollars to run a Monte Carlo simulation on a large language model (LLM), getting independent real-valued observations $X_1, \dots, X_N \stackrel{\text{i.i.d.}}{\sim} P$ for N very large. He estimates $\mu(P) = \mathbb{E}_P[X_i]$ via

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i,$$

getting a meaningfully positive value of $\bar{X}$. This is an exciting result because there is a consensus among philosophers of mind that $\mu(P)$ precisely quantifies the degree to which the LLM experiences true consciousness.

Just one problem: he only recorded $\bar{X}$ and deleted the data. Now a pesky referee is asking him for error bars!

Let $\sigma^2(P) = \text{Var}_P(X_i)$ denote the variance of a single observation. Assume throughout the problem that $\sigma^2(P) < \infty$. Let $\tau^2 = \frac{\sigma^2}{N}$ denote the variance of $\bar{X}$.

You tell him don't worry, he can just re-run his estimator on a new, smaller sample $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} P$, to get error bars.

We'll consider an asymptotic regime where $N \rightarrow \infty$ and the small sample size $n(N) \rightarrow \infty$ as well, but possibly not at the same rate (i.e., it could be that $n(N)/N \rightarrow 0$). If you apply asymptotic theorems from class, be sure to say which ones you use at each step.

- (a) Give a finite-sample unbiased estimator $\hat{\tau}^2(Y)$, based on Y only, for τ^2 . Show that it is unbiased, and that it is consistent in the sense that $\hat{\tau}^2/\tau^2 \xrightarrow{P} 1$.

Solution:

Define the sample variance $S_Y^2 = \frac{1}{n-1} \sum_i (Y_i - \bar{Y})^2$, where $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$. Then

$$\begin{aligned} (n-1)S_Y^2 &= \sum_i (Y_i^2 - 2Y_i\bar{Y} + \bar{Y}^2) \\ &= \left(\sum_i Y_i^2 \right) - n\bar{Y}^2. \end{aligned}$$

We have $\mathbb{E}_P[Y_i^2] = \mu^2 + \sigma^2$ and likewise $\mathbb{E}_P[\bar{Y}^2] = \mu^2 + \sigma^2/n$ since $\bar{Y} \sim (\mu, \sigma^2/n)$. Hence the expectation of the whole expression is $n(\mu^2 + \sigma^2) - n(\mu^2 + \sigma^2/n) = (n-1)\sigma^2$. As a result,

$$\mathbb{E}_P[S_Y^2/N] = \sigma^2/N = \tau^2.$$

Moreover,

$$\frac{n-1}{n} S_Y^2 = \left(\frac{1}{n} \sum_i Y_i^2 \right) - \bar{Y}^2 \xrightarrow{P} (\mu^2 + \sigma^2) - \mu^2 = \sigma^2,$$

giving $S_Y^2 \xrightarrow{P} \sigma^2$ since $\frac{n-1}{n} \rightarrow 1$. And $\hat{\tau}^2/\tau^2 = S_Y^2/\sigma^2 \xrightarrow{P} 1$.

(b) The referee is asking for a confidence interval too. Define

$$C(\bar{X}, Y) = [\bar{X} - \hat{\tau} z_{\alpha/2}, \bar{X} + \hat{\tau} z_{\alpha/2}],$$

where $\hat{\tau} = \sqrt{\hat{\tau}^2}$ and $z_{\alpha/2}$ is the upper $\alpha/2$ quantile of a $N(0, 1)$ random variable. Show that the coverage probability of C converges to $1 - \alpha$ asymptotically.

Solution:

The coverage event is:

$$\mu \in C \iff |\bar{X} - \mu| \leq \hat{\tau} z_{\alpha/2} \iff \frac{|\bar{X} - \mu|}{\hat{\tau}} \leq z_{\alpha/2}$$

From our previous result, $\hat{\tau}^2/\tau^2 \xrightarrow{P} 1$, so we also have $\hat{\tau}/\tau \xrightarrow{P} 1$, by the continuous mapping theorem. Since $\frac{\bar{X} - \mu}{\tau} \Rightarrow N(0, 1)$ by the CLT, we have by Slutsky that

$$R_N = \frac{\bar{X} - \mu}{\hat{\tau}} = \frac{(\bar{X} - \mu)/\tau}{\hat{\tau}/\tau} \Rightarrow N(0, 1)$$

Since the limiting distribution is continuous, all quantiles converge to normal quantiles, so

$$\begin{aligned} \mathbb{P}(|R_N| \leq z_{\alpha/2}) &= F_{R_N}(z_{\alpha/2}) - F_{R_N}(-z_{\alpha/2}) \\ &\rightarrow \Phi(z_{\alpha/2}) - \Phi(-z_{\alpha/2}) \\ &= 1 - \alpha. \end{aligned}$$

(c) Assume (for this part **only**) that P is $N(\mu, \sigma^2)$, for unknown μ and σ^2 . Show that if we replace the quantile $z_{\alpha/2}$ with a quantile $c_{\alpha/2}$ of some other distribution, we'll get an *exact* confidence interval in finite samples. What distribution should we use?

Solution:

The coverage event is the same, only now we have $Z = \frac{\bar{X} - \mu}{\tau} \sim N(0, 1)$, and $V = (n - 1)\hat{\tau}^2/\tau^2 = (n - 1)S_Y^2/\sigma^2 \sim \frac{1}{n-1}\chi_{n-1}^2$, independently. As a result,

$$R_N = \frac{Z}{\sqrt{V/(n-1)}} \sim t_{n-1},$$

so $\mathbb{P}(|R_N| < c_{\alpha/2}) = 1 - \alpha$, if $c_{\alpha/2}$ is the upper $\alpha/2$ quantile of the Student's t distribution with $n - 1$ degrees of freedom.

- (d) We can actually do a little better than $\bar{X}$ now that we have the additional data. Give an unbiased estimator based on $(\bar{X}, Y_1, \dots, Y_n)$ with smaller variance than $\bar{X}$, and give its variance. For full credit, you should give the *best* such estimator (smallest variance) but you don't need to show it is the best.

Solution:

If we had the entire data set, including the data that was deleted, the UMVU nonparametric estimator would be

$$\hat{\mu} = \frac{1}{N+n} \left(\sum_{i=1}^N X_i + \sum_{i=1}^n Y_i \right) = \frac{N\bar{X} + n\bar{Y}}{N+n}.$$

This estimator can be calculated from the available data $(\bar{X}, Y)$, so we are not going to do any better.

2. Poisson with additive offsets (30 points, 5 points / part).

Some useful facts for this problem:

- The $\text{Pois}(\lambda)$ distribution for $\lambda > 0$ has probability mass function

$$\frac{\lambda^x e^{-\lambda}}{x!}, \quad \text{for } x = 0, 1, \dots$$

It has mean λ and variance λ .

- If $X_i \stackrel{\text{ind.}}{\sim} \text{Pois}(\lambda_i)$ then $\sum_i X_i \sim \text{Pois}(\sum_i \lambda_i)$.
- For $w = (w_1, \dots, w_n)$ and $\lambda = (\lambda_1, \dots, \lambda_n)$, with all $w_i, \lambda_i > 0$, let $Q_{\lambda, w}$ denote the distribution of $\sum_i w_i X_i$, where $X_i \stackrel{\text{ind.}}{\sim} \text{Pois}(\lambda_i)$.

Assume we observe $X_i \stackrel{\text{ind.}}{\sim} \text{Pois}(a_i + \theta)$ for $i = 1, \dots, n$, with **known offsets** $a_1, \dots, a_n > 0$, and unknown parameter $\theta > -\min_i a_i$.

In both parts below, where I ask you to give a test, it is good enough to (1) define a test statistic and (2) define the rejection threshold in terms of the upper α quantile of a specific distribution. Don't worry about randomizing at the boundary.

(a) Consider the estimator

$$\delta(X) = \frac{1}{n} \sum_i (X_i - a_i).$$

Show that it is unbiased and give its variance.

Solution:

The expectation of $\delta(X)$ is

$$\frac{1}{n} \sum_{i=1}^n (\mathbb{E}_\theta X_i - a_i) = \frac{1}{n} \sum_{i=1}^n \theta = \theta,$$

and (by independence) the variance is

$$\frac{1}{n^2} \sum_{i=1}^n \text{Var}_\theta(X_i) = \frac{\sum_{i=1}^n (a_i + \theta)}{n^2}.$$

- (b) Show that $\delta(X)$ is UMVU in the equal-offset case where $a_1 = \dots = a_n = a$, for some value $a > 0$.

Solution:

In the equal-offset case, $X_i \stackrel{\text{i.i.d.}}{\sim} \text{Pois}(a + \theta)$. This is equivalent to the reparameterized family where $\lambda(\theta) = a + \theta > 0$, a full-rank one-parameter exponential family with sufficient statistic $T(X) = \sum_i X_i$. Then $\delta(X) = \frac{T}{n} - a$ is an unbiased estimator that is a function of the complete sufficient statistic $T(X)$; hence it is UMVU.

- (c) Now assume that not all of the a_i values are equal. Find the maximum likelihood estimator $\hat{\theta}_{\text{MLE}}$ for θ as a solution to an equation $f(\hat{\theta}; X_1, \dots, X_n, a_1, \dots, a_n, n) = 0$. (You do not need an explicit form for $\hat{\theta}$ but you should have an explicit form for f).

Solution:

The log-likelihood is

$$\ell_n(\theta; X, a) = \sum_{i=1}^n (\log(a_i + \theta) X_i - a_i - \theta - \log(X_i!)),$$

which is a concave function in θ . Differentiating gives

$$\dot{\ell}_n(\theta) = \sum_{i=1}^n \left(\frac{X_i}{a_i + \theta} - 1 \right),$$

so we are looking for a root of $f(\theta) = -n + \sum_{i=1}^n \frac{X_i}{a_i + \theta}$.

- (d) Define σ^2 as the asymptotic variance of $\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta)$. Show that σ^2/n is no larger than the variance of $\delta(X)$, and strictly smaller in the unequal-offset case.

Don't worry about the fact that the X_i values are not i.i.d.; just assume the MLE has the asymptotic distribution prescribed by the usual theory.

Hint: Note the Cauchy-Schwarz inequality implies that

$$\left(\sum_i z_i \right) \left(\sum_i z_i^{-1} \right) \geq n^2 \quad \text{for } z_1, \dots, z_n > 0,$$

with equality if and only if all z_i are equal. You may use this result without proof.

Solution:

The full-sample Fisher information is

$$J_n(\theta) = \text{Var}(\dot{\ell}_n(\theta)) = \sum_{i=1}^n \frac{1}{(a_i + \theta)^2} \text{Var}_\theta(X_i) = \sum_i \frac{1}{a_i + \theta},$$

and applying our standard result we have

$$\hat{\theta}_{\text{MLE}} \approx N(\theta, J_n(\theta)^{-1}),$$

That is, $\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta) \Rightarrow N(0, \sigma^2)$, and the approximate variance of $\hat{\theta}_{\text{MLE}}$ is

$$\sigma^2/n = J_n(\theta) = \left(\sum_i z_i^{-1} \right)^{-1},$$

for $z_i = a_i + \theta$.

By contrast, the variance of $\delta(X)$ (as we have shown) is $\frac{1}{n^2} \sum_{i=1}^n z_i$. Applying the inequality, we have

$$\frac{\text{Var}_\theta(\delta(X))}{\sigma^2/n} = \frac{1}{n^2} \left(\sum_i z_i \right) \left(\sum_i z_i^{-1} \right) \geq 1,$$

with equality if and only if all z_i are equal; or equivalently, when all a_i are equal.

- (e) Suppose that we want to test $H_0 : \theta = 0$ vs $H_1 : \theta = \theta_1$, for a **specific value** $\theta_1 > 0$. What test should we use?

Solution:

We should use the Neyman–Pearson likelihood ratio test, which rejects for large values of

$$\begin{aligned} \log \frac{p_{\theta_1}(X)}{p_0(X)} &= \ell_n(\theta_1; X, a) - \ell_n(0; X, a) \\ &= \sum_{i=1}^n \left(\log \left(\frac{a_i + \theta_1}{a_i} \right) \cdot X_i \right) - n\theta_1. \end{aligned}$$

Equivalently it rejects for large values of $\sum_i w_i X_i$ for

$$w_i = \log \left(\frac{a_i + \theta_1}{a_i} \right) > 0.$$

The cutoff is the upper α quantile of $Q_{\lambda, w}$ for $\lambda_i = a_i$.

- (f) (*) Suppose we want to test $H_0 : \theta \leq 0$ vs $H_1 : \theta > 0$. Suggest a test that is valid in finite samples, and that is as powerful as possible against values of θ that are barely larger than 0. For full credit, explain why your test is valid in finite samples.

Solution:

If we take $a \downarrow 0$ then the log-likelihood ratio divided by a converges to the score at $\theta = 0$:

$$\dot{\ell}_n(0; X, a) = \sum_i \left(\frac{X_i}{a_i} - 1 \right),$$

giving the most powerful test for small a . Equivalently, the test rejects for large values of $\sum_i w_i X_i$, for $w_i = 1/a_i$, which has distribution $Q_{\lambda, w}$ for $\lambda_i = a_i$.

Moreover, the distribution of the test statistic $\dot{\ell}_n(0; X, a)$ is stochastically increasing in θ , since it is a sum of independent terms each of which is stochastically increasing in θ .

3. Inference on a discrete CDF (25 points, 5 points / part).

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} P$ be a random sample from an unknown distribution P supported on $\{1, 2, \dots, 9\}$. Define the cumulative distribution function

$$F(k) = \mathbb{P}(X_i \leq k), \quad k = 1, \dots, 9,$$

and the empirical CDF

$$\hat{F}_n(k) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i \leq k\}.$$

Note that $\hat{F}_n(9) = F(9) = 1$ deterministically. Throughout this problem, we therefore work with the vectors

$$\hat{\mathbf{F}} = (\hat{F}_n(1), \dots, \hat{F}_n(8)), \quad \mathbf{F} = (F(1), \dots, F(8)).$$

This problem asks you to come up with asymptotic tests and confidence regions. You do not need to prove they have the right asymptotic coverage, just give the right tests and regions in terms of data and quantiles of well-specified distributions.

(a) Show that

$$\sqrt{n}(\hat{\mathbf{F}} - \mathbf{F}) \Rightarrow N_8(0, \Sigma(\mathbf{F})),$$

and give an explicit expression for the covariance matrix $\Sigma(\mathbf{F})$. (Your answer should give $\Sigma_{k\ell}$ as a function of $F(k)$ and $F(\ell)$.)

Solution:

Defining $T_i(k) = \mathbf{1}\{X_i \leq k\}$, we have $\mathbb{E}T_i(k) = F(k)$, $\text{Var}T_i(k) = F(k)(1 - F(k))$, and for $j \leq k$,

$$\Sigma_{jk} = \Sigma_{kj} = \text{Cov}(T_i(j), T_i(k)) = \mathbb{E}T_i(j)T_i(k) - F(j)F(k) = F(j)(1 - F(k)).$$

Hence $(T_i(1), \dots, T_i(8)) \sim (\mathbf{F}, \Sigma)$, and by the multivariate CLT, we have

$$\sqrt{n}(\hat{\mathbf{F}} - \mathbf{F}) \Rightarrow N_8(0, \Sigma).$$

- (b) Using the asymptotic distribution from part (a), give an asymptotic level- α test of

$$H_0 : \mathbf{F} = \mathbf{F}_0 \quad \text{vs} \quad H_1 : \mathbf{F} \neq \mathbf{F}_0,$$

for a specific point null $\mathbf{F}_0$. You may assume without proof that $\Sigma(\mathbf{F}_0)$ is invertible.

Solution:

If we premultiply by $\Sigma^{-1/2}$ we get

$$\sqrt{n}\Sigma^{-1/2}(\mathbf{F}_0) \left(\widehat{\mathbf{F}} - \mathbf{F}_0 \right) \Rightarrow N_8(0, I_8),$$

and taking the squared norm gives

$$W^* = \frac{1}{n} \left\| \Sigma^{-1/2}(\mathbf{F}_0) \left(\widehat{\mathbf{F}} - \mathbf{F}_0 \right) \right\|_2^2 \Rightarrow \chi_8^2,$$

so we reject for W^* larger than the upper α quantile of a χ_8^2 distribution.

- (c) By inverting the test from part (b), give an asymptotic $(1 - \alpha)$ Wald confidence region for $\mathbf{F}$. Note the covariance matrix $\Sigma(\mathbf{F})$ is unknown.

Solution:

We just plug in $\widehat{\Sigma} = \Sigma(\widehat{\mathbf{F}}) \xrightarrow{P} \Sigma(\mathbf{F})$, by the continuous mapping theorem (since $\widehat{\mathbf{F}} \xrightarrow{P} \mathbf{F}$ by LLN). Then we also have, by Slutsky,

$$W = \frac{1}{n} \left\| \widehat{\Sigma}^{-1/2} \left(\widehat{\mathbf{F}} - \mathbf{F} \right) \right\|_2^2 \Rightarrow \chi_8^2$$

- (d) Using the marginal asymptotic distribution of each $\widehat{F}_n(k)$, construct a simultaneous $(1 - \alpha)$ confidence band for $F(1), \dots, F(8)$. (A conservative band is acceptable.)

Solution:

Using the univariate asymptotic distribution $\widehat{F}_n(k) \approx N(F(k), \sigma_k^2)$ for $\sigma_k^2 = \Sigma_{kk} = F(k)(1 - F(k))$, we can just use a Bonferroni correction to construct:

$$C_k = \left[\widehat{F}_n(k) - \hat{\sigma}_k z_{\alpha/16}, \widehat{F}_n(k) + \hat{\sigma}_k z_{\alpha/16} \right],$$

which achieves the correct coverage because

$$\mathbb{P}(F(k) \notin C_k) \rightarrow \alpha/8,$$

and therefore

$$\mathbb{P}(F(k) \notin C_k \text{ for some } k) \leq \sum_k \mathbb{P}(F(k) \notin C_k) \leq \alpha.$$

- (e) (*) The random variables $\hat{F}_n(1), \dots, \hat{F}_n(8)$ are positively correlated. Explain how the joint asymptotic normal distribution from part (a) can be used to obtain a strictly narrower simultaneous confidence band than the one in part (d). You do not need to give critical values explicitly, but you should specify clearly how they would be defined or approximated.

Solution:

The previous answer was very conservative. Thinking a bit harder about the construction, the event that simultaneous coverage *fails* is

$$\{F(k) \notin C_k \text{ for some } k\} = \left\{ \max_{k=1}^8 \frac{|F_n(k) - F(k)|}{\hat{\sigma}_k} > z_{\alpha/16} \right\}.$$

Define the root

$$R_n = \max_{k=1}^8 \frac{|F_n(k) - F(k)|}{\hat{\Sigma}_{kk}} = \left\| \text{diag}(\hat{\sigma}_1^2, \dots, \hat{\sigma}_8^2)^{-1/2} (\hat{\mathbf{F}} - \mathbf{F}) \right\|_{\infty}.$$

If we use intervals $C_k = \hat{F}_n(k) \pm \hat{\Sigma}_{kk} c$, then the non-coverage event is exactly $R_n > c$. Previously we used a crude union bound here, but we could instead do it by bootstrapping X_i values or by simulating

$$R_n^* = \left\| \text{diag}(\hat{\sigma}_1^2, \dots, \hat{\sigma}_8^2)^{-1/2} (\hat{\mathbf{F}}^* - \mathbf{F}) \right\|_{\infty}$$

where $\hat{\mathbf{F}}^*$ is drawn from the approximate multivariate Gaussian model $N_8(\hat{\mathbf{F}}, \hat{\Sigma})$.

We can simulate repeatedly from this model to obtain an estimate of the upper α quantile c_{α} . This will generally be much shorter due to the high correlation between adjacent entries of $\hat{\mathbf{F}}$.

4. Ising model (25 points, 5 points / part).

Consider the one-parameter family of distributions on $\{0, 1\}^d$ (with $d \geq 2$), given by

$$p_\beta(x) = \frac{1}{Z(\beta)} \prod_{i < j} e^{\beta |x_i - x_j|},$$

where the normalizing constant is

$$Z(\beta) = \sum_{x \in \{0, 1\}^d} \prod_{i < j} e^{\beta |x_i - x_j|}.$$

This is called the *Ising model*. Let $S(x) = \sum_{i=1}^d x_i$ denote the sum of $x \in \{0, 1\}^d$. For $x \in \{0, 1\}^d$, let $S(x) = \sum_{i=1}^d x_i$ and $T(x) = S(x)(d - S(x))$.

- (a) Show that $\{p_\beta : \beta \in \mathbb{R}\}$ is a one-parameter exponential family and show that $T(X)$ is a complete sufficient statistic.

Solution:

We have

$$p_\beta(x) = \exp \left\{ \beta \sum_{i < j} |x_i - x_j| - A(\beta) \right\},$$

which is a full rank exponential family with sufficient statistic $T(X) = \sum_{i < j} |x_i - x_j|$. Moreover, if $S(X) = s$, then there are s values with $x_i = 1$ and $d - s$ with $x_i = 0$. The total number of disagreements is the total number of pairs we can make by matching the 0s with the 1s, which is exactly $s(d - s)$. Hence $T(X) = S(X)(d - S(X))$.

- (b) Is S a sufficient statistic? Is it a complete sufficient statistic? Justify your answers.

Solution:

S is sufficient because T is sufficient, and we can calculate T from S . However, S is not *complete* sufficient because both S and $d - S$ are equally likely (and they map to the same T). So if we take $g(S) = S - d/2$ as our function, we have $\mathbb{E}_\beta g(S) \neq 0$ for all β , but $g(S)$ is not almost surely zero.

- (c) Let $A(\beta) = \log Z(\beta)$ and consider the prior family on β given by

$$\lambda_{\eta, \kappa}(\beta) \propto \exp\{\eta \beta - \kappa A(\beta)\}, \quad \kappa > 0.$$

Show that this family is conjugate for the model, and give the parameters $(\eta_{\text{post}}, \kappa_{\text{post}})$ for the posterior after observing a single draw $X \sim p_\beta$.

Solution:

We obtain the posterior by multiplying the prior and the likelihood:

$$\begin{aligned}\lambda(\beta | X) &\propto \exp\{\eta\beta - \kappa A(\beta)\} \cdot \exp\{\beta T(X) - A(\beta)\} \\ &\propto \exp\{(\eta + T)\beta - (1 + \kappa)A(\beta)\} \\ &\propto \lambda_{\eta+T, \kappa+1}(\beta)\end{aligned}$$

(d) Give a level- α UMP test of

$$H_0 : \beta \leq 0 \quad \text{vs} \quad H_1 : \beta > 0.$$

It is enough to (i) specify a test statistic, and (ii) define the rejection threshold in terms of a quantile of a precisely specified distribution. (You do not need to randomize at the boundary.)

Solution:

This family has monotone likelihood ratios in $T(X) = S(X)(d - S(X))$, so we reject when T is large, or equivalently when $|S - d/2|$ is small. Under the null, $S \sim \text{Binom}(d, 1/2)$, so we will reject if

$$c_{(1-\alpha)/2} < S - d/2 < c_{(1+\alpha)/2},$$

which captures α mass in the middle of the binomial distribution. Or equivalently when $|S - d/2| < c_{(1+\alpha)/2}$.

(e) (*) Now introduce a nuisance parameter $\alpha \in \mathbb{R}$ and consider the two-parameter family

$$p_{\beta, \alpha}(x) \propto \exp\{\beta T(x) + \alpha S(x)\}, \quad x \in \{0, 1\}^d.$$

Suppose we observe i.i.d. samples $X^{(1)}, \dots, X^{(n)} \sim p_{\beta, \alpha}$. Give an *exact* level- α test of

$$H_0 : \beta \leq 0 \quad \text{vs} \quad H_1 : \beta > 0$$

that remains valid for all values of the nuisance parameter α . (Again it is enough to specify a test statistic and a rejection threshold via a quantile of a precisely specified distribution.)

Solution:

The multiple sample version has sufficient statistics $T_+ = \sum_i T_i$ and $S_+ = \sum_i S_i$. The usual UMPU prescription is to condition on $S_+ = \sum_{ij} X_j^{(i)}$ and reject for large values of T_+ . To get the conditional distribution, note that under the null all nd of these Bernoulli variables are i.i.d. and we are just conditioning on the total number of 1's. So the conditional distribution is obtained by permuting the cells in the $n \times d$ data matrix. We do this B times to get permuted data matrices $X^{*1}, \dots, X^{*B}$ and get the p -value

$$p(X) = \frac{1}{B+1} \left(1 + \sum_{b=1}^B 1 \left\{ T_+(X) \geq T_+(X^{*b}) \right\} \right)$$